

Comparative Efficacy of ICSI Versus Conventional IVF in Non-Male Factor Infertility

Kenisya Saravenen | BSc Biomedical Sciences | University of York

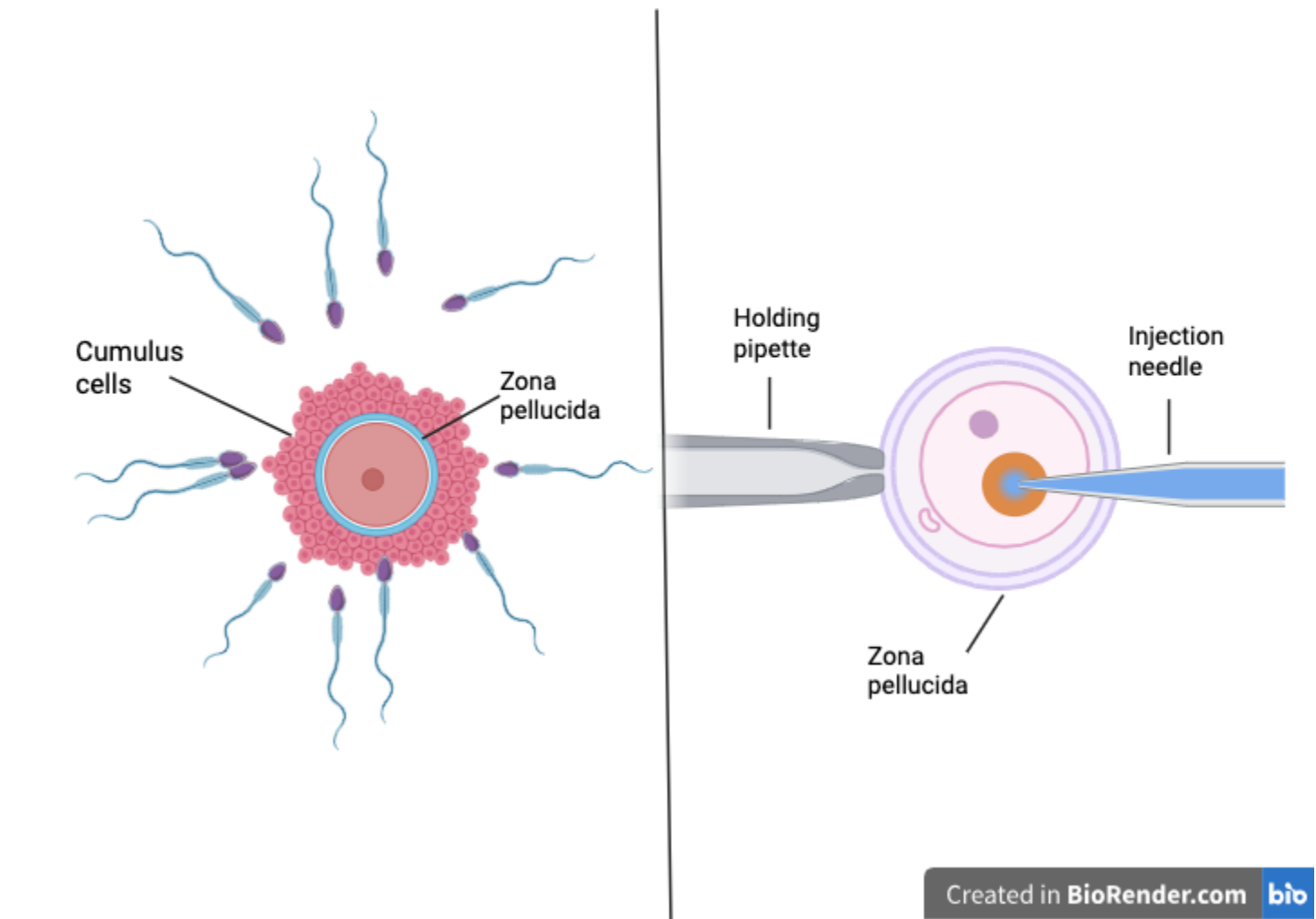
LITERATURE-BASED EVIDENCE SYNTHESIS • 2026

Should ICSI be the only method used and should IVF be scrapped?

01 BACKGROUND & BIOLOGICAL MECHANISMS

ASSISTED REPRODUCTIVE TECHNOLOGY

Conventional IVF (cIVF) and intracytoplasmic sperm injection (ICSI) both enable fertilisation outside the body, but they differ in how much of the sperm-oocyte interaction is bypassed.



cIVF ICSI

Conventional IVF

- Sperm-oocyte interaction is retained.
- Selection includes motility, zona pellucida binding and the acrosome reaction.

Intracytoplasmic sperm injection

- One sperm is immobilised and injected directly into the oocyte.
- Bypasses zona pellucida binding and acrosome-dependent penetration.

Biological trade-off

ICSI can overcome impaired sperm-oocyte interaction but also removes several natural selection steps. This micromanipulation artificially bypasses highly conserved pre-fertilization checkpoints, including: chemotactic transit, requisite capacitation and biochemical filtration inherent to cumulus oophorus and zona pellucida penetration. Advanced sperm-selection approaches aim to restore some of this selection before injection, but no single method is established as a gold standard.

FROM LABORATORY ENDPOINTS TO CLINICAL OUTCOMES



A higher fertilisation rate is an intermediate endpoint. The clinically important comparison is whether it improves pregnancy and live-birth outcomes.

02 CLINICAL INDICATIONS & PRACTICE

WHEN DOES ICSI HAVE A CLEAR ROLE?

Scenario	Clinical context	Implication
Severe male factor	Low sperm count / poor motility / poor morphology	ICSI can bypass impaired fertilisation steps
Surgically retrieved sperm	Obstructive or severe sperm-production problems	Direct injection facilitates use of retrieved sperm
Previous poor fertilisation	Previous IVF with none or very few eggs fertilised	ICSI may be used in a subsequent cycle
Non-severe / normal semen	No severe male-factor indication	Current evidence does not show better live-birth outcomes

THE “INDICATION CREEP” QUESTION

ICSI began as a treatment for severe male infertility but is now used beyond that group. The clinical question is whether a higher technical fertilisation rate translates into more embryos, pregnancies or live births when sperm quality is not severely impaired.

WHY MIGHT CLINICIANS STILL CHOOSE ICSI?

01 Avoiding total fertilisation failure

especially where a single treatment opportunity is important

02 Fertility preservation

ICSI may generate more fertilised eggs/embryos, but not all may be used

03 Laboratory preference

a deliberate, controlled micromanipulation step rather than sperm-oocyte co-incubation

Current guidance direction

HFEA states there is currently no scientific evidence supporting ICSI for infertility not related to sperm problems; the 2026 ASRM/SART committee opinion also focuses on where evidence supports or does not support non-male-factor indications.

PRACTICAL TREATMENT LOGIC

SEVERE MALE FACTOR ICSI

NO SEVERE MALE FACTOR cIVF

Exception: previous total/near-total fertilisation failure may justify a different strategy. The choice remains patient- and clinic-specific.

03 EVIDENCE & STATISTICAL OUTCOMES

WHAT DO RANDOMISED STUDIES SHOW?

2024 multicentre RCT

2,387

couples randomised • Wang et al.

2025 multicentre RCT

824

women randomised • INVICSI

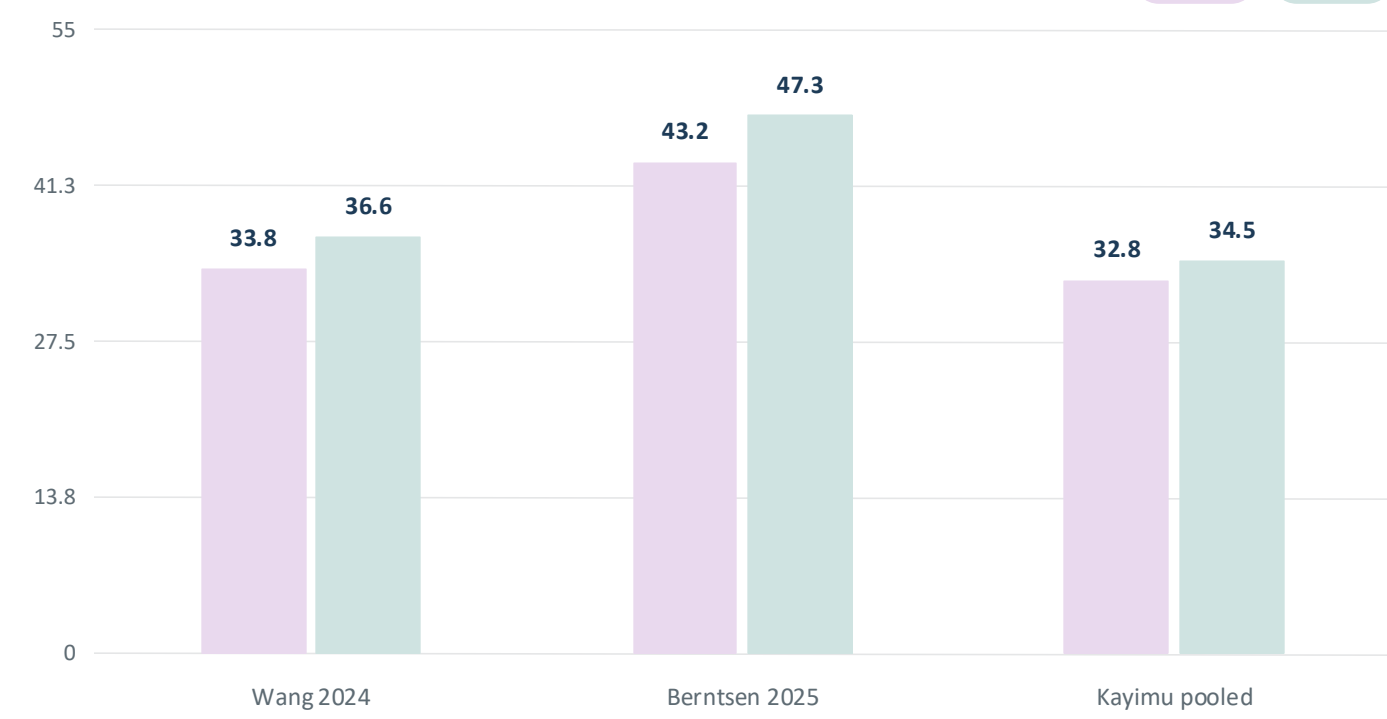
2026 meta-analysis

6 RCTs

without severe male-factor infertility

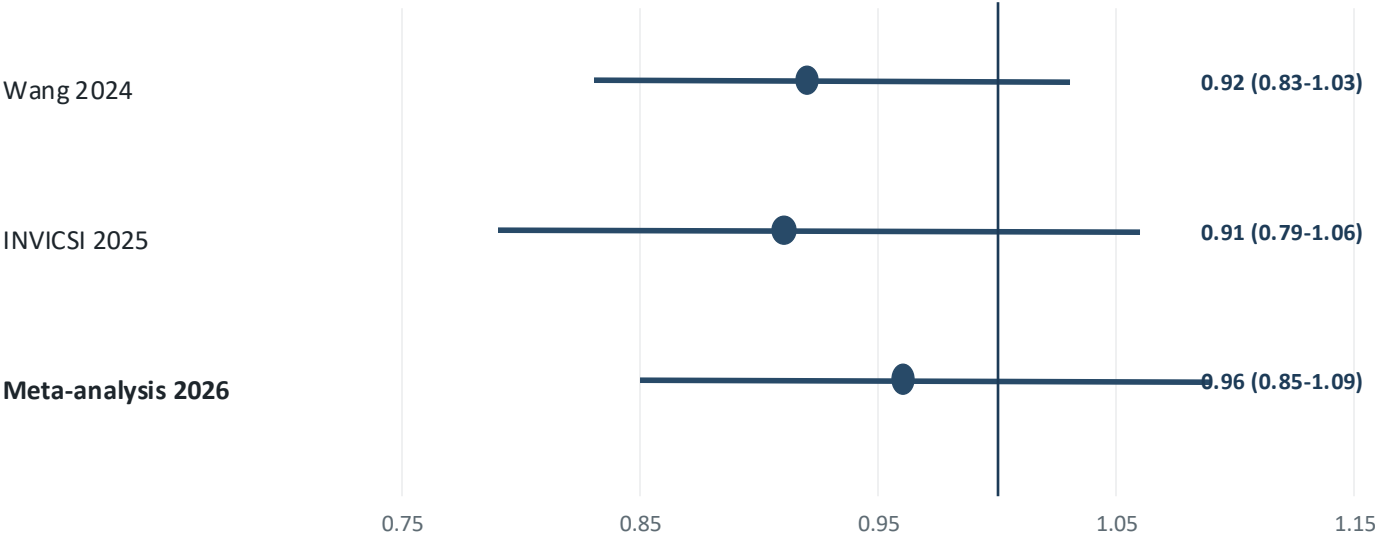
Study	Outcome	ICSI vs cIVF	Effect estimate
Wang 2024	First-transfer live birth	33.8% vs 36.6%	RR 0.92; p=0.16
Berntsen 2025	Cumulative live birth	43.2% vs 47.3%	RR 0.91 (0.79-1.06)
Kayimu 2026	Pooled live birth	32.8% vs 34.5%	RR 0.96 (0.85-1.09)
Kayimu 2026	Pooled cumulative LB	43.2% vs 47.4%	RR 0.92 (0.84-1.01)

Live birth outcomes reported in recent RCT evidence



Reported live-birth effect estimates

Risk ratio (RR); line at 1.0 = no difference



INTERPRETATION

Across recent randomised evidence, confidence intervals for the reported live-birth effects cross RR = 1.0. The 2026 meta-analysis rated the evidence for live birth and cumulative live birth as high quality.

EVIDENCE LIMITATIONS

- Only a limited number of high-quality RCTs report live-birth outcomes.
- Outcome definitions and patient populations vary across trials.
- Findings should not be extrapolated to severe male-factor infertility, where ICSI has a clear role.

04 CLINICAL CONSIDERATIONS & CONCLUSION

BENEFIT-RISK BALANCE

FERTILISATION

Technical advantage ≠ clinical advantage

ICSI can increase the number of fertilised eggs in some settings, but recent RCTs do not show higher live-birth rates without severe male-factor infertility.

PROCEDURE

More invasive micromanipulation

ICSI requires individual sperm selection and injection; HFEA notes a risk of egg damage during the procedure.

OFFSPRING HEALTH

Long-term causality remains uncertain

HFEA notes possible associations with genetic/developmental outcomes but says evidence cannot yet separate treatment effects from underlying infertility in all cases.

LABORATORY & COST

Additional specialist handling

Routine use outside evidence-based indications increases laboratory workload and procedure-specific cost without demonstrated live-birth benefit.

FERTILITY PRESERVATION - A USEFUL NUANCE

A 2021 analysis using HFEA fertility-preservation data reported more fertilised eggs/embryos with ICSI, but 19% of eggs were not used in ICSI treatment. The authors concluded technique choice should depend on sperm characteristics rather than applying ICSI universally.

CONCLUSION

For infertility without severe male-factor indications, recent randomised and pooled evidence does not show a live-birth advantage for ICSI over conventional IVF. ICSI remains an important treatment when sperm-related fertilisation impairment is substantial, while routine use outside those indications should be evidence-led.

SELECTED REFERENCES

- Wang Y, et al. Lancet. 2024;403:924-934. doi:10.1016/S0140-6736(23)02416-9.
- Berntsen S, et al. Nat Med. 2025;31:1939-1948. doi:10.1038/s41591-025-03621-x.
- Kayimu K, et al. Hum Reprod. 2026;41:1173-1182. doi:10.1093/humrep/deag066.
- ASRM/SART Practice Committee. Fertil Steril. 2026;126:49-56. doi:10.1016/j.fertnstert.2026.03.032.
- Aggarwal B, et al. Reprod Fertil. 2021;2:1:1-13. doi:10.1530/humrep-20-0059.
- Baldini D, et al. Cells. 2021;10:3566. doi:10.3390/cells10123566.
- HFEA. Intracytoplasmic sperm injection (ICSI) and fertility treatment guidance. Accessed 2026.